SUTRA-TRACE

NEURO-SYMBOLIC DIGITAL FORENSICS FRAMEWORK

Multi-Source Evidence Correlation & Court-Admissible Incident Reconstruction · Where every fragmented trace connects with cryptographic certainty

SUTRAM 2026 NATIONAL HACKATHON

IGDTUW · MeitY ISEA-III · NFSU Lead Institute

ISO/IEC 27037 Aligned

BSA 2023 · Section 63 Ready

Runs on Standard LEA Workstations

Problem Statement ID	PS-01 · Building AI-Powered Digital Forensics Investigation Framework
Theme	AI-Powered Cyber Investigation, Incident Response & Digital Forensics Technologies
Team Name (registered on portal)	SUTRA-Trace

Disk Images · Memory Dumps · Network PCAPs · WhatsApp Telemetry · UPI Ledger ■■[Cryptographic Provenance Rail]■■> Court Dossier

02 TEAM DETAILS

A team may have up to 4 members. Fill in only the cards you need and delete the rest.

01 TEAM LEADER

Full Name: _____

Branch / Year: _____

Email: _____

Phone: _____

02 MEMBER

Full Name: _____

Branch / Year: _____

Email: _____

Phone: _____

03 MEMBER (if applicable)

Full Name: _____

Branch / Year: _____

Email: _____

Phone: _____

04 MEMBER (if applicable)

Full Name: _____

Branch / Year: _____

Email: _____

Phone: _____

IDEA TITLE

WHAT TO COVER

- Detailed explanation of the proposed solution
- How it addresses the problem
- Innovation and uniqueness of the solution

PROPOSED SOLUTION — DESCRIBE YOUR IDEA / SOLUTION / PROTOTYPE

SUTRA-Trace: AI finds the lead, cryptography proves it, BSA 2023 certifies it

■ Evidence Explosion & Silos

100+ GB per case: disk images, network PCAPs, memory dumps, UPI statements, each trapped in separate tools.

■ AI Hallucination Trap

Generic AI models output unanchored probabilistic guesses that courts reject under Section 45 BSA.

■ Indian Threat Sophistication

Digital Arrest deepfake calls, multi-tier UPI mule smurfing, and sideloaded trojan APKs defeat siloed suites.

① Ingestion & Vault

Read-only intake hashed on arrival (SHA-256 + 1 KB BLAKE3 Merkle-DAG).

② Neural Triage

AI flags suspicious LOLBins, synthetic deepfakes, and UPI mule patterns as leads.

③ Provenance Graph

Links multi-source artifacts into one time-ordered attack storyline (T-PKG).

④ Section 63 Dossier

Auto-generates Schedule Certificate drafts (Part A & B) with hash manifest.

Neuro-Symbolic: No timeline claim without a hash-linked raw evidence p

India-First: BSA/BNSS 2023, UPI, I4C Threat Focus

Openly Verifiable with standard hash tools

INVESTIGATIVE VELOCITY: 72+ hours of manual correlation ■■■> ~15-minute verifiable court dossier (target)

WHAT TO COVER

- **Technologies to be used (e.g. programming languages, frameworks, hardware)**
- Methodology and process for implementation (flow charts / images / working prototype)

YOUR TECHNICAL APPROACH — PART 1 · INGESTION & MULTI-MODAL TRIAGE

① INGESTION & INTEGRITY ■■> ② NEURAL TRIAGE ■■> [③ T-PKG Correlation ■■> ④ Legal Dossier (Next Page →)]

① INGESTION & INTEGRITY LAYER (SYMBOLIC)

- **Read-Only Block Ingestion:** ISO/IEC 27037 compliant mounting with dual-bus eBPF write-blocking (PCIe NVMe 0x09 + SCSI 0x42) and APST=0 BGC quiescing.
- **Linear SHA-256 + Piecewise Hash:** Piecewise Valid-Extent SHA-256 over healthy sector bitmask M survives physical media degradation over trial lifespan.
- **Native 1 KB BLAKE3 Merkle Tree:** Domain-separated bitflags (0x01..0x08) guarantee 100% root parity with b3sum CLI live tools.
- **Sec. 105 BNSS Seizure Video Bind:** Parses MP4 stts/ctts VFR curves and tracks 50 Hz power-grid hum phase ($\Delta\phi < 0.08$ rad) to verify recording continuity.
- **Supported Forensic Inputs:** Raw Disk (RAW/E01), Volatile RAM, PCAP, EVTX/Sysmon, WhatsApp logs, and UPI bank CSVs.

② MULTI-MODAL NEURAL TRIAGE (NEURAL LEADS)

- **Log Telemetry (Fine-Tuned SecBERT):** Detects stealth Living-Off-The-Land binaries (certutil, powershell -enc, mshta) in EVTX/Sysmon logs.
- **Synthetic Media / Deepfake Triage:** Combines QP-gated 2D-DCT with Carotid-to-Supraorbital PTT pulse transit lag ($\Delta\tau \in [12\text{ms}, 45\text{ms}]$) to expose face swaps.
- **Financial Mule Layering Detector:** Multi-horizon flow conservation across {3m, 15m, 2h, 24h} windows flags pass-through mule chains and Soundbox laundering.
- **Dual-Path WhatsApp & VoIP Telemetry:** Carves plaintext ICE candidate logs (srflx/prflx IPs) from whatsapp.log on victim phones for Sec. 94 BNSS IPDR notices.
- **Core Rule:** AI output functions strictly as an investigative lead, never as a stand-alone court finding.

ENTERPRISE OPEN TECHNOLOGY STACK:

Systems & Acquisition:	Rust · Go · libtsk · Scapy · Volatility 3 · blake3 SIMD
AI / ML Runtimes:	PyTorch · Hugging Face (SecBERT) · ONNX Runtime INT8 (7-Bit)
Backend & Forensic Notary:	Python 3.11 · ReportLab (PDF/A-1b) · Cryptography · SQLite WAL

TECHNICAL APPROACH (CONTD.)

WHAT TO COVER

- Technologies to be used (e.g. programming languages, frameworks, hardware)
- **Methodology and process for implementation (flow charts / images / working prototype)**

YOUR TECHNICAL APPROACH — PART 2 · T-PKG GRAPH & COURTROOM ADMISSIBILITY

③ TEMPORAL PROVENANCE KNOWLEDGE GRAPH (T-PKG)

- Entities: Process ■■> Network Socket ■■> IP/Domain ■■> File Hash ■■> UPI VPA.
- 32-Byte Contiguous CSR Arena: Holds 4,000,000 active edges in 128 MB RAM (<210 MB tota
- Bijective Harmonic Time: [cos■sin■ln(1+Δt)] eliminates periodic time aliasing.
- 5-Invariant Anti-Timestomping: NTFS MFT LSN (0x08) & $SI USN (0x28) monotonicity.
- Deterministic PCST Solver: Bit-identical 25-node minimal causal courtroom storylines.

④ STATUTORY LEGAL DOSSIER ENGINE (BSA 2023)

- Timeline built strictly from evidence-pinned edges (source hash + byte offset).
- Section 63 BSA 2023 Schedule: Part A (Custodian) & Part B (Forensic Expert).
- 3-Scenario Custody Bridge: Victim Part A, Seized Device as Sec. 57 Expl. 5 Primary Evidence.
- Section 105 BNSS Cross-Binding: Seizure video hash sealed into master Merkle root.
- Section 45 BSA Exculpatory Ledger: Attached append-only audit trail of all pruned telemetry.

5-STEP INVESTIGATION METHODOLOGY:

1. Ingest & Hash (ISO 27037) ■■> 2. Neural Triage ■■> 3. Graph Correlation ■■> 4. Investigator Review (HITL) ■■> 5. Legal Dossier

WORKING PROTOTYPE ARTIFACTS (DEMO READY IN REPOSITORY):

- **Investigator Cockpit:** sector_e_dashboard.py: Live web cockpit with 1-click 'Operation Maya' Digital Arrest Case.
- **Evidence Vault:** sector_a_vault.py: ISO 27037 dual-hashing, Merkle tree, and BNSS 105 seizure video binder.
- **Official Schedule Cert:** BSA_2023_Sec63_Certificate.pdf: Print-ready 2-page legal certificate with scan-ready QR code.
- **Architecture Flowchart:** https://share.gemini.google/VGvgu3LxFA4E (Clickable interactive diagram link)

FEASIBILITY AND VIABILITY

WHAT TO COVER

- **Analysis of the feasibility of the idea**
- Potential challenges and risks
- Strategies for overcoming these challenges

YOUR FEASIBILITY ANALYSIS — PART 1 · BUILT FOR A 16 GB POLICE LAPTOP

16 GB RAM	7-Bit INT8	0 BYTES
Standard LEA workstation profile (zero GPU dependency)	Quantized ONNX models with zero CPU overflow	Written to source evidence (strict read-only)

TECHNICAL FEASIBILITY

- CPU Inference via ONNX Runtime: Models quantized to 7-bit symmetric INT8 ([-64, +63]) with Q16.16 lookup tables.
- Streaming Chunked I/O: 4 MB sliding mmap window with explicit MADV_DONTNEED eviction prevents page-cache exhaustion.
- Sliding Time Windows: KAIROS 15-minute graphlets compress inactive entities into WL HistoSketches (<210 MB total RAM).

OPERATIONAL FEASIBILITY

- Air-Gapped Operation: 100% offline edge execution with zero cloud telemetry; evidence never leaves Indian soil.
- Strict ISO/IEC 27037 Compliance: Zero-alteration verified via dual SHA-256 and native 1 KB BLAKE3 Merkle-DAG roots.
- Interoperable: Ingests raw dd, E01, PCAP, EVTX, and Volatility memory dumps without replacing existing lab tools.

ECONOMIC & ORGANIZATIONAL VIABILITY

- Open-Source Architecture: Completely eliminates expensive ■15–25 lakh proprietary forensic suite licenses.
- Modular Design: Ingestion, Triage, Graph, and Legal modules update independently as threats evolve.
- Low Training Overhead: Streamlined Cockpit interface automates drafting, reducing examiner cognitive load by >80%.

FEASIBILITY AND VIABILITY (CONTD.)

WHAT TO COVER

- Analysis of the feasibility of the idea
- **Potential challenges and risks**
- **Strategies for overcoming these challenges**

YOUR FEASIBILITY ANALYSIS — PART 2 · RISKS AND MITIGATIONS

RISK / ADVERSARIAL CHALLENGE	IMPACT	STRATEGIC ENGINEERING MITIGATION
Tampered Timestamps on Files	**High**	5-Invariant Anti-Timestomping: NTFS MFT LSN (0x08) & $SI USN (0x28) monotonicity + carved USN Journal V2/V3 and VSS baselines.
AI False Positives / Hallucination	**High**	Deterministic Provenance Pinning: every timeline claim points to physical storage offsets; unpinned outputs remain advisory leads.
Large Data Volumes / Out-of-Memory	**Medium**	Double-buffered 4 MB direct-I/O with MADV_DONTNEED + KAIROS 15-minute sliding-window graphlets keep heap strictly < 210 MB.
Deepfakes Evolving / Generator Drift	**Medium**	Tri-modal bio-physics fusion: Carotid PTT phase delay + VAE upsampling lattice residuals catch generative models regardless of visual realism.
Courtroom Cross-Examination / AI Challenge	**High**	Evidence relies 100% on physical hex byte offsets and standard b3sum hashes that independent experts can re-verify without AI weights.

Human-in-the-Loop Sign-off | **Full Audit Log of Excluded Items (Section 45 BSA)** | **100% Reproducible b3sum Hashes**

06 IMPACT AND BENEFITS

WHAT TO COVER

- **Potential impact on the target audience**
- **Benefits of the solution (social, economic, environmental, etc.)**

YOUR IMPACT AND BENEFITS

ALIGNED WITH I4C & MHA PRIORITY THREATS: Digital Arrest Extortion · Multi-Tier UPI Mule Networks · Sideloaded APK Fraud

LAW ENFORCEMENT & CYBER CELLS

~95% LESS

CORRELATION TIME (TARGET)

• Rapid Triage Acceleration: Evidence triage from 72+ hours to ~15 minutes on 250 GB images.
• Dedicated Playbooks: Built-in correlation for Digital Arrest, UPI smurfing, and APK fraud.
• Unified Investigative Console: Ingests all sources into one pane instead of 5+ isolated tools.

JUDICIARY & PROSECUTION

100%

HASH-LINKED TO EVIDENCE

• Unshakeable Evidentiary Admissibility: Built natively around Section 63 BSA 2023 Schedule drafts.
• Explainable Courtroom Storylines: Translates hex bytes into clear, chronological visual narratives.
• Independent Verification: Any forensic expert can re-verify byte offsets with standard open tools.

ATMANIRBHAR BHARAT

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LICENCE AVOIDED / SEAT (EST.)

• Indigenous Self-Reliance: Open alternative to costly foreign suites (Cellebrite, EnCase, Magnet).
• Complete Data Sovereignty: Sensitive sovereign crime telemetry never leaves Indian borders.
• Adaptive to Indian Threat Patterns: Built specifically for NPCI UPI, Indian languages, and BSA.

BENEFIT PILLARS: Social: Faster relief for fraud victims · Economic: Reduced lab licensing costs · Environmental: Runs on existing PCs with zero GPU farms

RESEARCH AND REFERENCES

WHAT TO COVER

- **Details / links of the reference and research work**

YOUR RESEARCH AND REFERENCES

STATUTORY & STANDARDS

- **Bharatiya Sakshya Adhiniyam, 2023:** Sections 61, 62, 63 & The Schedule (Part A & B mandatory certificates); Section 57 (Explanation 5); Section 45.
- **Bharatiya Nagarik Suraksha Sanhita, 2023:** Section 105 (Mandatory audio-video recording of search & seizure) & Section 94 (Summons to produce electronic documents).
- **ISO/IEC 27037:2012 Standard:** Guidelines for identification, collection, acquisition, and preservation of digital evidence.
- **NIST SP 800-86 & RFC 3161:** Integrating forensic techniques into incident response & cryptographic Time-Stamp Protocol.

ACADEMIC LITERATURE (NFSU)

- **Prof. (Dr.) Parag H. Rughani (NFSU):** Dean, SCSDF — Foundational research on machine learning in digital forensics, Smart Acquisition, and Smart Reporting.
- **SynFO: Forensic Triage (2023):** Gogia & Rughani — High-concurrency disk imaging and ML triage combining Go/Rust systems with Python AI.
- **FSAGen & PAREX (NFSU):** Prof. Parag Rughani — File system artifact generation for tool testing & robust corrupted partition parsing.
- **Provenance Attack Reconstruction:** Top-tier cybersecurity research in IEEE S&P, ACM CCS, and NDSS on causal graph reduction.

BENCHMARK DATASETS

- **NIST CFReDS Reference Sets:** Computer Forensic Reference Data Sets for validating NTFS MFT, registry timelines, and hash tools.
- **DARPA Transparent Computing:** Engagement 3 & 5 (CADETS, THEIA ground-truth audit benchmarks for provenance graphs).
- **CIC CSE-CIC-IDS2018 (AWS):** Comprehensive network flow dataset for multi-hop intrusion and network graph validation.
- **FaceForensics++ (ICCV 2019):** Benchmark dataset for synthetic face and video deepfake detection under compression.

SUTRA-Trace: Every fragmented trace, connected with cryptographic certainty.